

WEEKLY REPORT :

Project Title: Self Driving RC Car

Date: 9/22/2026

Instructor: Dr. Nguyen

TA: Ruben Ramirez

Project Members:

- Ethan Wells (Group Leader)
- Alexis Perez
- Ethan Bishop
- Abigail Duran

Weekly Summary (9/22/2026):

- This week the team moved from selecting parts to testing them on hardware. Ethan Wells set the machine-learning goals aside to get the compute ready for the rest of the team: he assembled the Raspberry Pi Compute Module 5 on the Compute Module 5 IO Board with its heatsink and PWM fan, CR2032 real-time-clock battery, and 27 W USB-C power supply, flashed Raspberry Pi OS Lite (64-bit) to the 128 GB NVMe solid-state drive that the board boots from, and brought the board up headless over SSH through a USB Wi-Fi dongle, which the CM5 needs because it has no onboard wireless; he traced very slow SSH throughput to the dongle's stock driver and replaced it with a working one, and wrote setup.sh so any team member can install the shared software prerequisites on a board with one command. Alexis Perez finalized the ultrasonic sensor implementation on the Pi, which measures distance to obstacles and walls, and tested the TT-style drive motors and L298N motor controller, finding performance issues that may require replacement parts; he also determined the hardware power requirements that will drive battery selection. Ethan Bishop ordered 2WD and 4WD Raspberry Pi HAT motor controllers and wrote code to test them for operational accuracy, and continued implementing the IMU, the sensor that reports the vehicle's heading and turns, verifying its measurements and directions. Abigail Duran prepared the libraries and software for the LCD1602

RGB status display, researched camera setup and configuration for the CM5, and identified the steps needed to integrate the camera and display with the machine-learning model.

Proposed Plan for Next Week:

- Ethan Wells: Prepare the LISA dataset (US traffic signs) for training and test the trained German-sign (GTSRB) model.
- Alexis Perez: Select replacement motors and motor controller if the tested parts do not meet requirements, and narrow down the battery and power-supply options.
- Ethan Bishop: Finish testing the Pi HAT motor controllers, finish the IMU implementation on the Raspberry Pi, and research 3D-printable STL files for the chassis.
- Abigail Duran: Continue hardware testing and verify communication between the Raspberry Pi and all connected components, determine which components and connections go on the final PCB, and begin the PCB layout including component placement, board dimensions, power distribution, and signal routing.

Weekly Contributions:

- Ethan Wells:
- 9/19/2026 – Assembled the CM5 on the Compute Module 5 IO Board with the heatsink and PWM fan, CR2032 RTC battery, and 27 W USB-C power supply, then flashed Raspberry Pi OS Lite (64-bit) to the 128 GB NVMe SSD and booted the board from M.2 (3 hours)
- 9/21/2026 – Brought up headless SSH over the USB Wi-Fi dongle and, after tracing very slow SSH throughput to the dongle's stock driver, replaced it with a working driver (2 hours)
- 9/23/2026 – Wrote setup.sh, a one-command install of the shared software prerequisites (git, python3, pip, venv), and updated the parts list with the remaining compute purchases (2 hours)
- Alexis Perez:
- Finalized the ultrasonic sensor implementation on the Raspberry Pi, including additional distance-reading and range testing

- Tested the TT-style drive motors and L298N motor controller for performance and identified issues with the current components
- Determined the hardware power requirements to guide battery and power-supply selection
- Ethan Bishop:
- Ordered 2WD and 4WD Raspberry Pi HAT motor controllers and wrote code to test them for operational accuracy
- Continued implementing the IMU on the Raspberry Pi and verified its measurements and directions
- Abigail Duran:
- Prepared the necessary libraries and software for the LCD1602 RGB display module
- Researched camera setup and configuration for the Raspberry Pi Compute Module 5
- Identified the steps needed to integrate the camera and LCD with the machine-learning model
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Hour Tracker:

Name	Hours For the Week	Cumulative Hours
Ethan Wells	7	19
Alexis Perez	7	16
Ethan Bishop	6	15
Abigail Duran	6	17